

Effects of Mandated Maternity Leave on Young Women's Labor Market Outcomes

Pulak Ghosh, Stephanie Hao, Lisa Ho, Garima Sharma and Shreya Tandon

January 8, 2025

Motivation

- Over 140 countries offer paid maternity leave to young mothers (World Bank 2020)
 - ▶ India (2017), Nigeria (2018), Pakistan (2020)

Motivation

- Over 140 countries offer paid maternity leave to young mothers (World Bank 2020)
 - ▶ India (2017), Nigeria (2018), Pakistan (2020)
- Impact on mothers, children
 - ▶ Lalive & Zweimuller (2009), Dustmann & Schönberg (2012), Schönberg & Ludsteck (2014), Bailey, Byker, Patel & Ramnath (2019), Rossin-Slater (2011), and more
- This paper: Effects on young women in anticipation of motherhood
 - ▶ Wages, employment, careers

This Paper: Effects of 2017 Maternity Leave Law in India

- Increased duration of paid maternity leave from 12 to 26 weeks

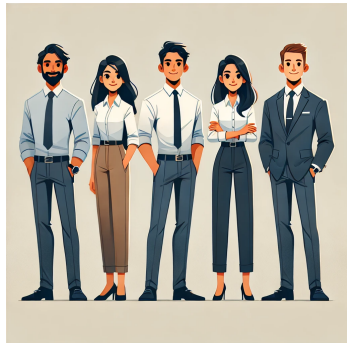
This Paper: Effects of 2017 Maternity Leave Law in India

Blue Collar Workers



- $\leq$ Rs.15,000 (727 USD PPP)
- Lose worker for 14 additional weeks
- Government pays salary

White Collar Workers



- $>$ Rs.15,000
- Lose worker for 14 additional weeks
- Employer pays salary

DiD Empirical Strategy

- DiD comparing establishments already offering 26 weeks (control) to those offering 12 weeks (treated)
- Parent company policies
- Data: Glassdoor reviews, newspaper announcements, ongoing survey of representative sample of 1200 firms

DiD Empirical Strategy

- DiD comparing establishments already offering 26 weeks (control) to those offering 12 weeks (treated)
- Parent company policies
- Data: Glassdoor reviews, newspaper announcements, ongoing survey of representative sample of 1200 firms
- Control: 7% of employment
 - ▶ Same industry mix
 - ▶ Slightly larger (160 vs. 84)

DiD specification

(1) Control for size differences:

$$Y_{jt} = \sum_{t=-4}^{t=7} \beta_t Treat_j 1_t + \alpha_j + \lambda_{sbt} + \epsilon_{jt}$$

- j = establishment, t = month relative to April 2017
- $\lambda_{sbt} = \text{State} \times \text{Size-quartile-bin} \times \text{Time fixed effects}$

(2) Report effects for large firms (> 100 workers)

Identifying assumption: parallel evolution

Blue Collar Workers



- Employer-employee linked social security records (EPFO)
- Wage, employment, 2014 - 2018

White Collar Workers



- LinkedIn profiles (universe)
- Education, Careers, 2012 - 2023

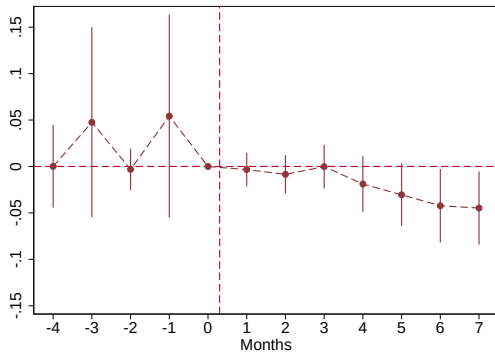
→ Coverage: 55 million workers (40 million EPFO, 15 million LinkedIn)

→ 41% of non-agricultural workforce

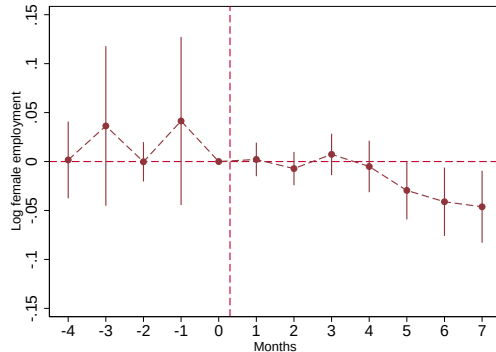
Blue Collar Workers (social security records)

Decreases women's employment by 6% within six months

Large firms (>100 workers)



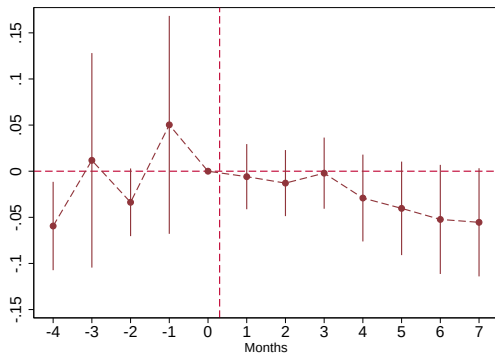
All firms



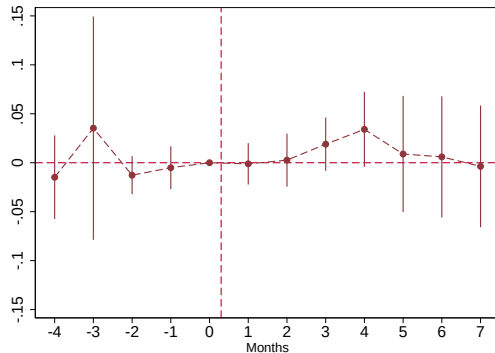
(80% of employment)

Effects concentrated among young women (18 - 35 years)

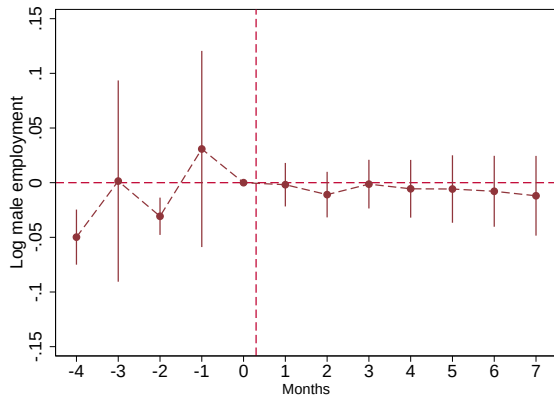
Young (18-35 years)



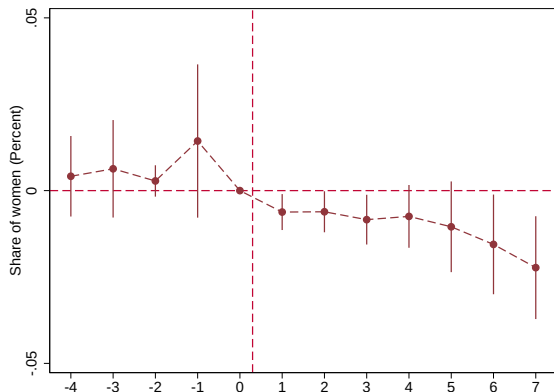
Over 35 years



No effect on male employment



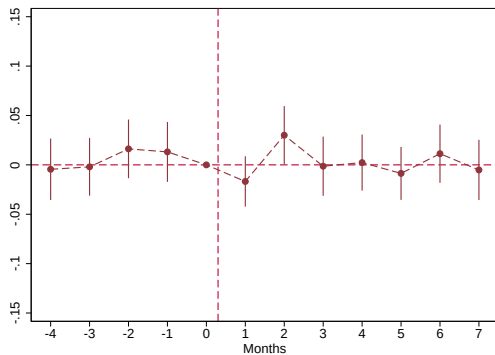
Share of women declines by 2%



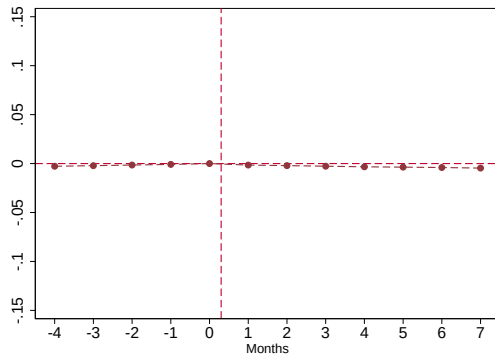
- Loss of 250,000 jobs within six months
- Decomposition: 55% from incumbents, 45% from reduced hiring

Contract work? (No)

Female share, contract firms



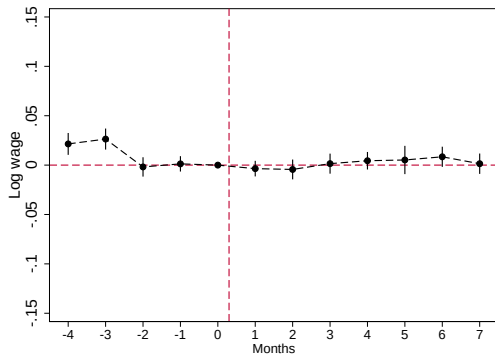
Female incumbent, switch to contract firm



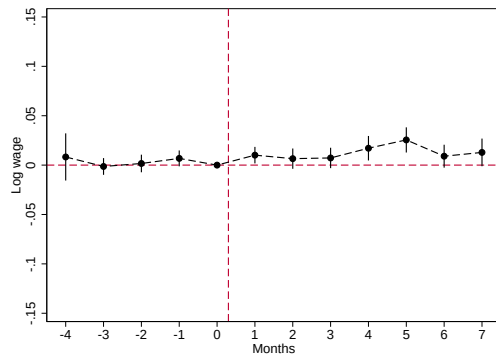
- Top 30 contract firms in India ($\sim 1/3$ of workers)
- Not switching to informality

Wages: no effect for women, small increase for men

Female incumbents



Male incumbents



- Magnitude, men: 1.6%

No effect on new workers' wages

	Women	Men
Treat*Post	-0.013 (0.008)	-0.013 (0.009)
Observations	389,143	389,143

Summary: Effects of 2017 Maternity Benefit Law on Women's Labor Market outcomes

- Employment: Reduced female employment by 6% within 6 months (250,000 jobs)
- Effects concentrated among young women (18 - 35 years)
- Wages: No impact on women's wages, increased male wages by 1.6%

Magnitudes: Employer misperceptions

To justify the magnitude of employment decline, employers must greatly overestimate rate of maternity leave taking

Magnitudes: Employer misperceptions

To justify the magnitude of employment decline, employers must greatly overestimate rate of maternity leave taking

Quick calculation:

$$\text{Hire woman if: } B > p * c$$

where B = benefit of employing woman, p = probability of maternity leave, c = cost

Magnitudes: Employer misperceptions

To justify the magnitude of employment decline, employers must greatly overestimate rate of maternity leave taking

Quick calculation:

$$\text{Hire woman if: } B > p * c$$

where B = benefit of employing woman, p = probability of maternity leave, c = cost

Assume:

$$\Delta c = 14 * \text{salary}, \quad B \sim U(\text{mean} = 0.18 * \text{salary})$$

Given:

$$\Delta \log(f) = -0.06$$

Magnitudes: Employer misperceptions

To justify the magnitude of employment decline, employers must greatly overestimate rate of maternity leave taking

Quick calculation:

$$\text{Hire woman if: } B > p * c$$

where B = benefit of employing woman, p = probability of maternity leave, c = cost

Assume:

$$\Delta c = 14 * \text{salary}, \quad B \sim U(\text{mean} = 0.18 * \text{salary})$$

Given:

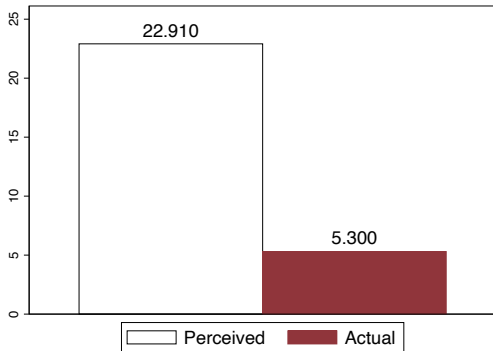
$$\Delta \log(f) = -0.06$$

→ **Perceived probability of maternity leave in first year** $\sim 27\%$. **Actual** $\sim 5\%$ [Details](#)

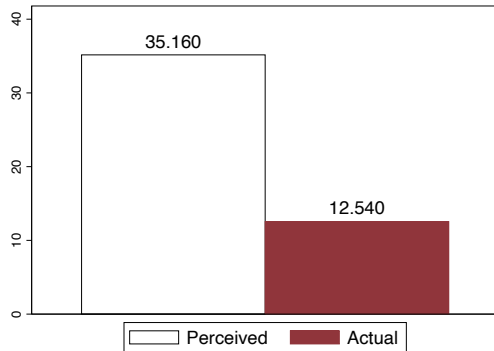
Survey evidence of misperceptions

- Pilot survey of 40 HR managers at Indian IT firms (Conlon & Sharma 2024)

Year 0-1: Perceived vs. Actual



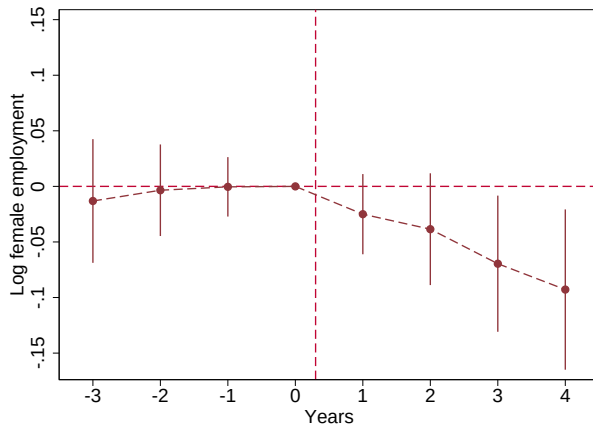
Year 0-2: Perceived vs. Actual



- Ongoing: Can correcting misperceptions reduce discrimination? [Question](#)

White Collar Workers (LinkedIn)

Decreases women's employment by 10% in four years

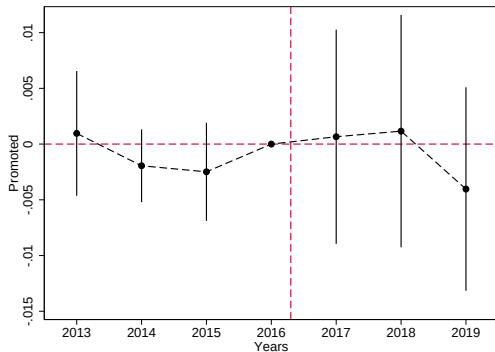


- $t = 1$ in 2017

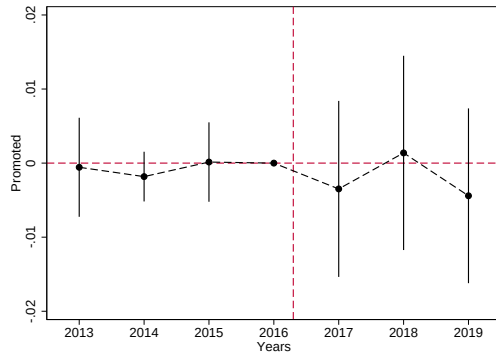
Career: Managerial positions

Career: Managerial positions

Female incumbents



Male incumbents



• + No effect on number of promotions.

Coming up: Career trajectories

Coming up: Career trajectories

- “Career clones”: Compare women to historic counterparts graduating from the same university + degree + similar experience
- Do women’s careers evolve differently from clones after vs. before the reform, relative to evolution for men?

Coming up: Career trajectories

- “Career clones”: Compare women to historic counterparts graduating from the same university + degree + similar experience
- Do women’s careers evolve differently from clones after vs. before the reform, relative to evolution for men?

Outcomes:

- First job: Time to find, firm quality
- Firm-specific HK - abstract, manual, routine roles
- Promotions
- Others?

Conclusion

- Maternity leave expansion in India, from 12 to 26 weeks, reduced female employment by 6 - 8% in short-run; 10% in four years
- Disproportionately impacted young women (18-35 years)
- Potentially reflects misperceptions
- Coming up: effect on women's career trajectories (occupations, promotions, job quality)

Policy

Thank you!

Misperceptions: Details

- Assume the 14 week expansion cost $\Delta c = 3.5 \text{ months} * \text{salary}$, although employers did not have to pay salary (Social security did for these workers, earning $< \text{Rs.15,000}$). Employers only lost the worker for 14 additional weeks.
- $B = mp - \text{wage}$. Assume 18% markdown/can pay equally productive woman 18pp less than man (Sharma 2023 monopsony estimate).
- $\Delta \log(f) = P[B > p * c_{\text{new}}] - P[B > p * c_{\text{old}}]$
- Use CDF of uniform distribution (+ new women's hiring declines by 0.22, incumbent women's employment declines 0.042)

Survey Question

At IT firms like yours in Tamil Nadu (between x_1 and x_2 employees, and with a similar turnover). For every 100 female employees that these firms hired in entry-level roles since 2016, how many women do you think:

- Took maternity leave within the first year?
- Took maternity leave between the first and second year?
- Did not take maternity leave within the first two years?

Policy implications

- Correcting Misperceptions 0.1cm
 - ▶ Win-win for worker, employer
- Cost-sharing
- Screening

[Back](#)